
Liquid Crystal Display
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Liquid Crystal Display
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Objectives

- Define the LCD
- Connect LCD in Arduino Projects
- Print text, sensor readings, and display special characters on the LCD

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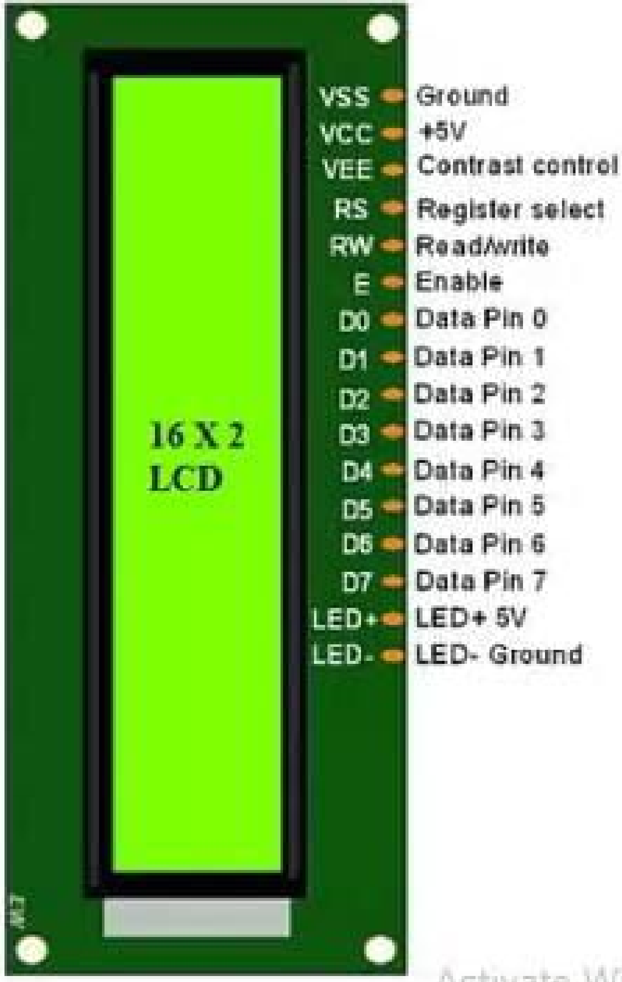
Liquid Crystal Display

LCD

LCD stands for Liquid Crystal Display.

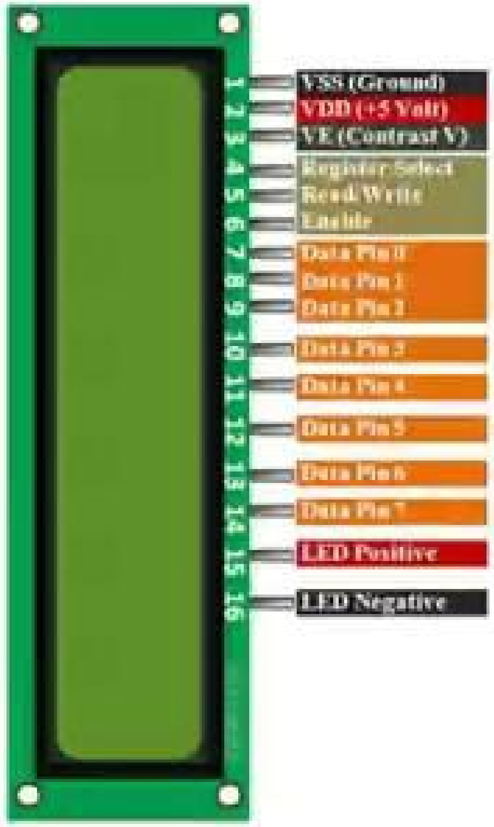
- An LCD is used to display text messages or present numbers from sensors. LCDs can be very handy in monitoring project.
- Liquid Crystal displays or LCDs have been used in electronics equipment since the late 1970s.
- LCD displays have the advantage of consuming very little current and they are ideal for your Arduino projects



LCD Pins

PINS	DETAILS	Power Pins
Pin1	VSS	It's a ground pin for common grounds.
Pin2	VDD	The power pin will use for voltage input to the 16X2 LCD.

PINS	DETAILS	Led Pins
Pin15	+ve	The LCD comes in multiple colors and every time in different LCD the color will depend on the internal LED. Pin15 is for the power input of the LED.
Pin16	-ve	Pin16 is the ground pin of the LED.



LCD

LCD stands for Liquid Crystal Display.

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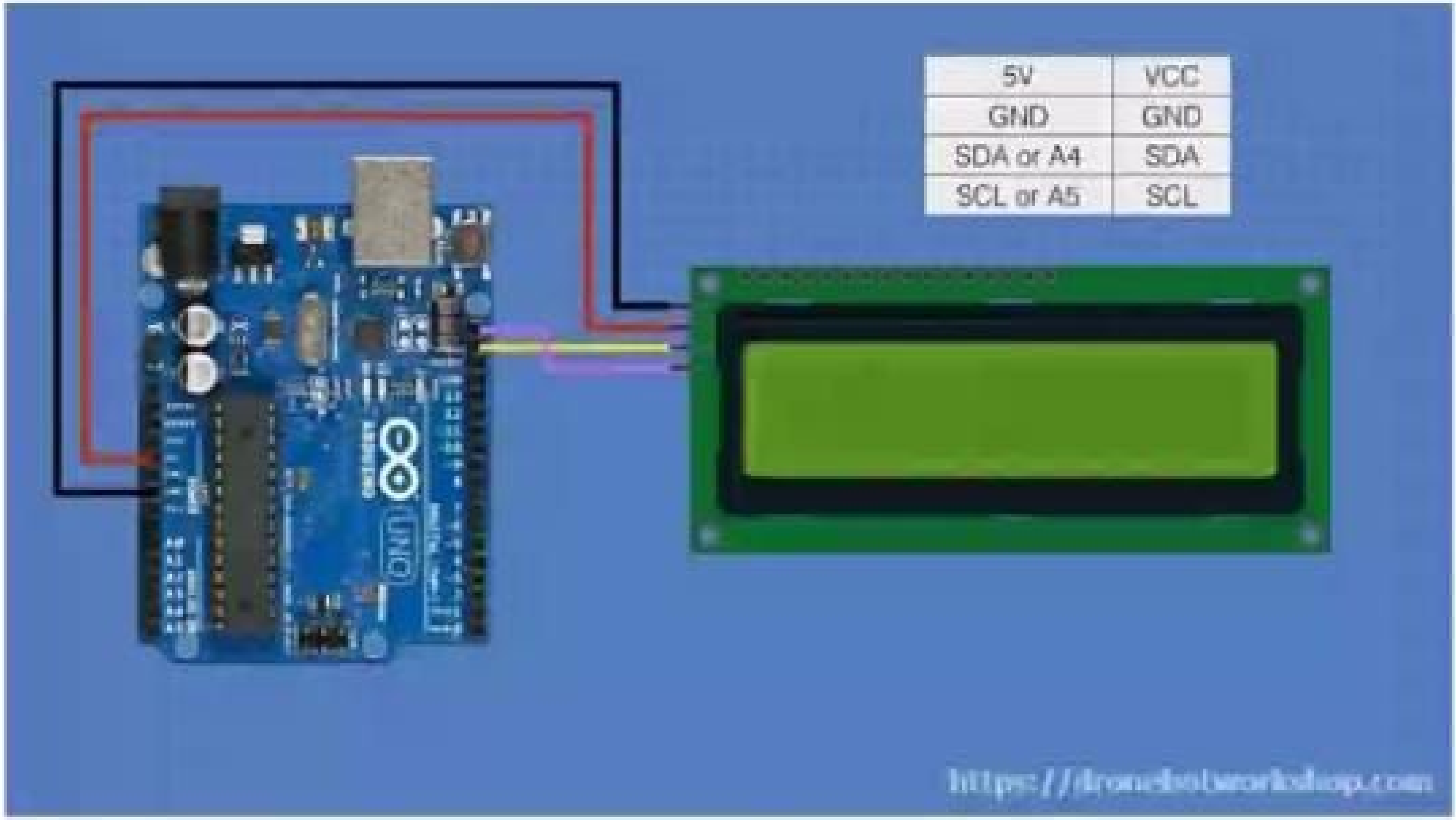
I2C Bus Adapter

The I2C bus adapter only uses four connections, two of them are for power:

1. **Power** – This can be either 5 Volts or 3.3 volts, depending upon the application. Note that there are many precautions that must be observed if you are interfacing a 3.3 volt and 5 volt I2C device on the same bus.
2. **Ground** – The ground connection.
3. **SDA** – Serial Data. This line is used for both transmit and receive.
4. **SCL** – Serial Clock. This is a timing signal supplied by the Bus Master device.

Board connection



Functions

Comment	Code Use	Code terms
LiquidCrystal lcd (rs, en, d4, d5, d6, d7);	This term is used to define the LCD. it defines the Arduino pins for the communication to be established	LiquidCrystal: command initiator lcd: Name of the LCD rs, en, d4, d5, d6, d7: digital port number responsible to communicate with RS, E, D4, D5, D6, and D7 on the LCD
lcd.begin(16, 2);	This term defines the LCD dimensions (length and width).	lcd: name of the LCD begin: command initiator (column, row)
lcd.print();	Printing on the LCD such that each character will be considered as a space.	("strings") ('char') (variables)

Functions

Comment	Code Use	Code terms
LiquidCrystal lcd (rs, en, d4, d5, d6, d7);	This term is used to define the LCD. it defines the Arduino pins for the communication to be established	LiquidCrystal: command initiator lcd: Name of the LCD rs, en, d4, d5, d6, d7: digital port number responsible to communicate with RS, E, D4, D5, D6, and D7 on the LCD
lcd.begin(16, 2);	This term defines the LCD dimensions (length and width).	lcd: name of the LCD begin: command initiator (column, row)
lcd.print();	Printing on the LCD such that each character will be considered as a space.	("strings") ('char') (variables)

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Functions

Comment	Code Use	Code terms
lcd.setCursor(,);	Set the cursor position	setCursor: command initiator (column, row)
lcd.createChar(1 ,heart);	This comment is used to form a char for LCD. the inserted array will be transformed into char	lcd: name of the LCD createChar command initiator (charNumber , arrayName)
lcd.write(1);	this comment is used to present custom characters	write: command initiator (1): the char number which should be present
lcd.blink();	Make the cursor space blink	Blink:command initiator
lcd.noBlink();	Stop the blinking of the cursor position.	noBlink: command initiator

Functions

Comment	Code Use	Code terms
lcd.cursor();	present the cursor on the LCD	cursor: command initiator
lcd.noCursor();	Hide the cursor on the LCD	noCursor: command initiator
lcd.autoscroll();	enable the auto scrolling on the LCD if we have a long word	autoscroll: command initiator
lcd.noAutoscroll();	disable auto scrolling	noAutoscroll: command initiator
lcd.display();	Turn ON the LCD display	display: command initiator
lcd.noDisplay();	Turn OFF the LCD display	noDisplay: command initiator
lcd.leftToRigh();	Move the LCD cursor from left to right when printing	leftToRight: command initiator

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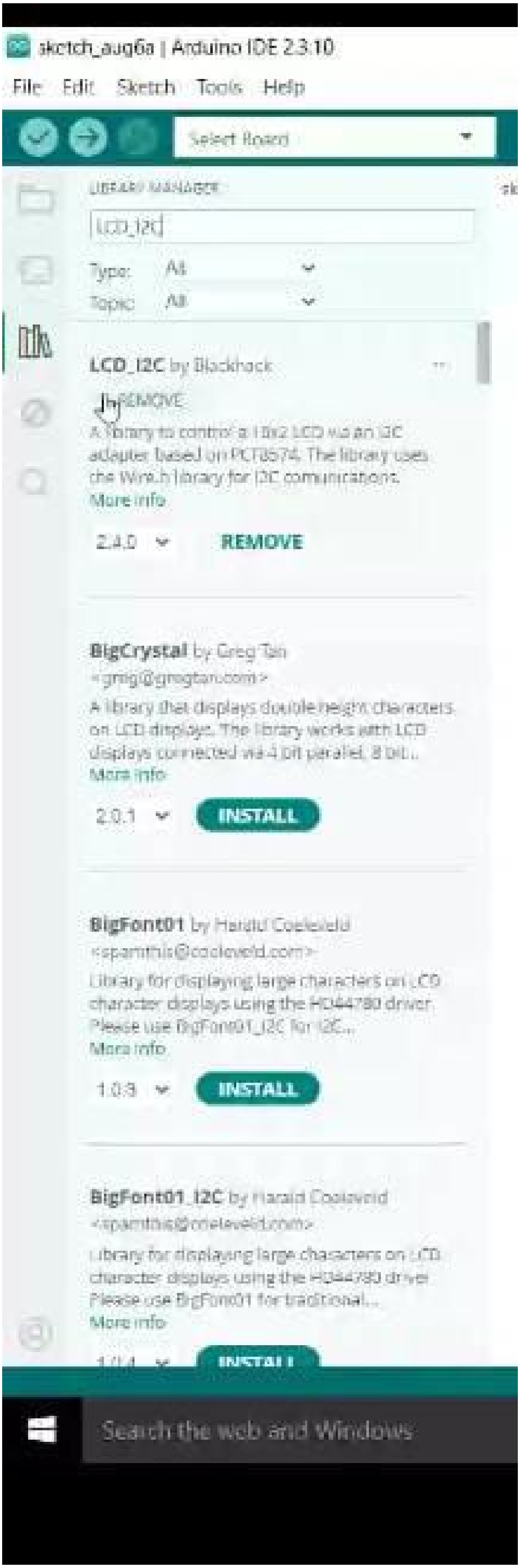
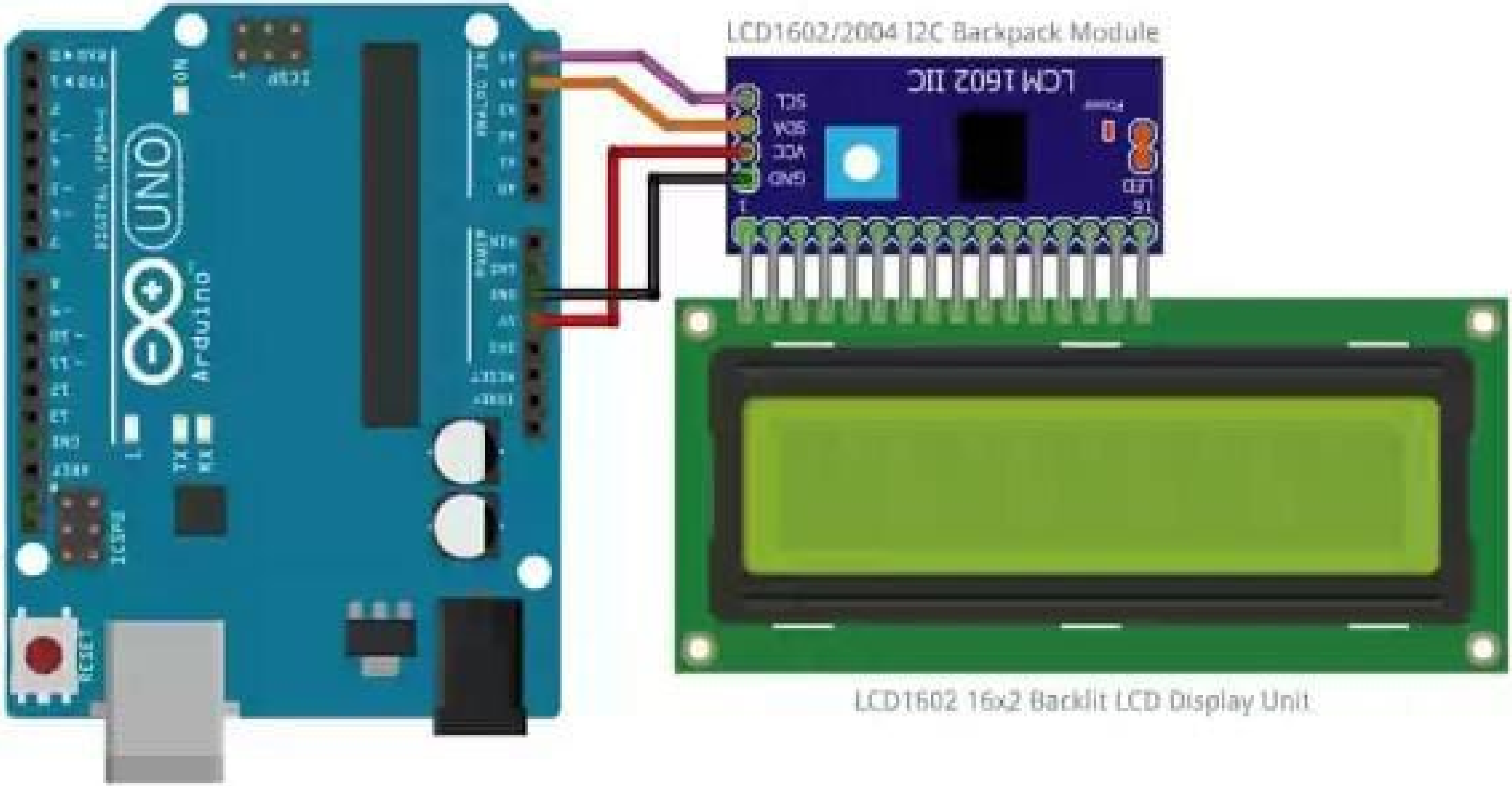
Functions

Comment	Code Use	Code terms
lcd.rightToLeft();	Move the LCD cursor from right to left when printing	rightToLeft: command initiator
lcd.home();	set the cursor on (0, 0) position	home: command initiator


sketch_aug6a | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Select Board
LIBRARY MANAGER
LCD_I2C
Type: All
Topic: All
LCD_I2C by Blackhack
A library to control a 16x2 LCD via an I2C adapter based on PCF8574. The library uses the Wire.h library for I2C comunications.
More info
2.4.0
REMOVE
BigCrystal by Greg Tan
A library that displays double height characters on LCD displays. The library works with LCD displays connected via 4 bit parallel, 8 bit...
More info
2.0.1
INSTALL
BigFont01 by Harald Coeleveld
Library for displaying large characters on LCD character displays using the HD44780 driver. Please use BigFont01_I2C for I2C...
More info
1.0.3
INSTALL
BigFont01_I2C by Harald Coeleveld
Library for displaying large characters on LCD character displays using the HD44780 driver. Please use BigFont01 for traditional...
More info
INSTALL
Search the web and Windows

Board connection



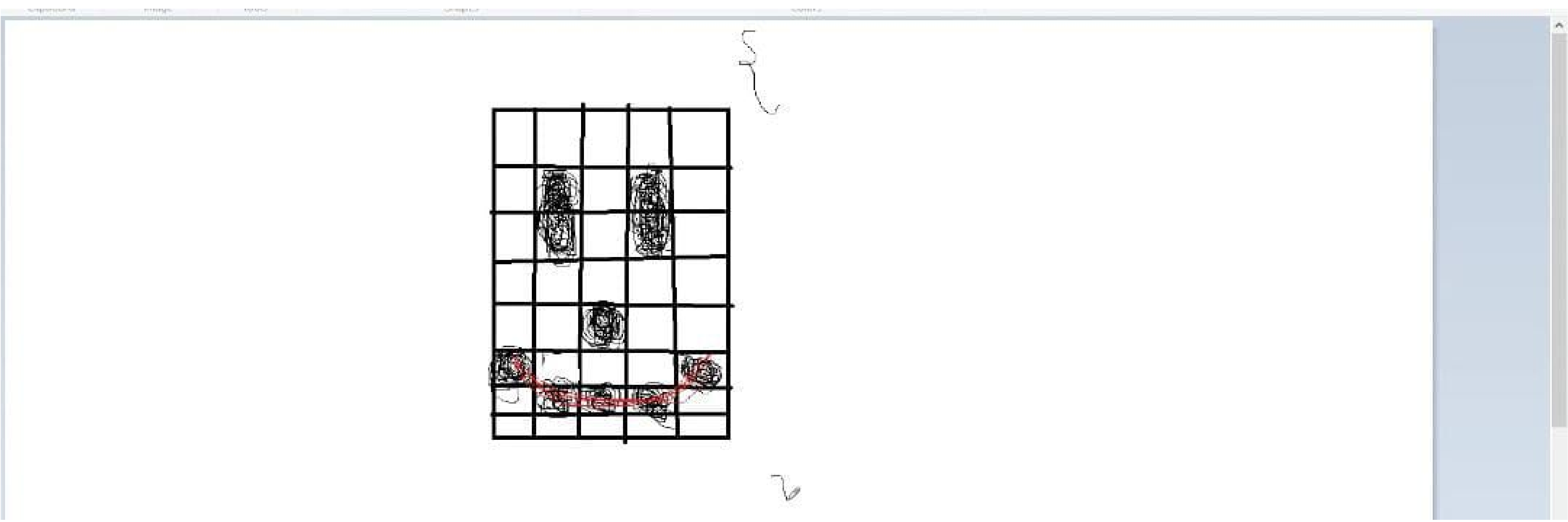
```
#include <Wire.h>
#include <LCD_I2C.h>

//Define the LCD
LCD_I2C lcd=LCD_I2C(0x27,16,2); //0x27 : code for using the I2C scanner ,16 : number of columns, 2: number of

void setup() {
  // put your setup code here, to run once:
lcd.begin(); // command to initialize the LCD
lcd.backlight(); //command to turn on the LCD Blue backlight
}

void loop() {
  // put your main code here, to run repeatedly:
lcd.setCursor(1,0); // command will set the cursor on the second column
}
```

sketch_aug6a.ino

```
#include <Wire.h>
#include <LCD_I2C.h>

//Define the LCD
LCD_I2C lcd=LCD_I2C(0x27,16,2); //0x27 : code for using the I2C scanner ,16 : number of columns, 2: number of

void setup() {
  // put your setup code here, to run once:
lcd.begin(); // command to initialize the LCD
lcd.backlight(); //command to turn on the LCD Blue backlight
}

void loop() {
  // put your main code here, to run repeatedly:
lcd.setCursor(1,0); // command will set the cursor on the second column (1) and the first row (0)
lcd.print("Hello Arduino !");
lcd.setCursor(2,1); //command will set the cursor on the third column (2) and the second row(1)
lcd.print("LCD Tutorial");
}

```


Output

```
Sketch uses 3326 bytes (10%) of program storage space. Maximum is 32256 bytes.
```



sketch_aug6d.ino

```
#include <SimpleDHT.h>
#include <LCD_I2C.h>
#include <Wire.h>

//DEFINE THE LED
LCD_I2C lcd=LCD_I2C(0x27,16,2);
//DEFINE THE DHT SENSOR
int pinDHT11=7;
Simple DHT11 dht11(pinDHT11);

void setup() {
  // put your setup code here, to run once:
lcd.begin();
lcd.backlight();

}

void loop() {
  // put your main code here, to run repeatedly:

}
```

Arduino Uno

sketch_aug6d.ino

```
int pinDHT11=7;
Simple DHT11 dht11(pinDHT11);

void setup() {
  // put your setup code here, to run once:
lcd.begin();
lcd.backlight();
}

void loop() {
  // put your main code here, to run repeatedly:
byte temperature;
byte humidity;
lcd.read(&temperature,&humidity,NULL);

lcd.setCursor(0,0);
lcd.print("Temperature:");
lcd.print(temperature);
lcd.print("°");  //alt 248

lcd.setCursor(0,1)

```

Activate Windows
Go to Settings to activate Windows.

Ln 28, Col 19   Arduino Uno on COM3 [not connected]

File Edit Sketch Tools Help

Arduino Uno

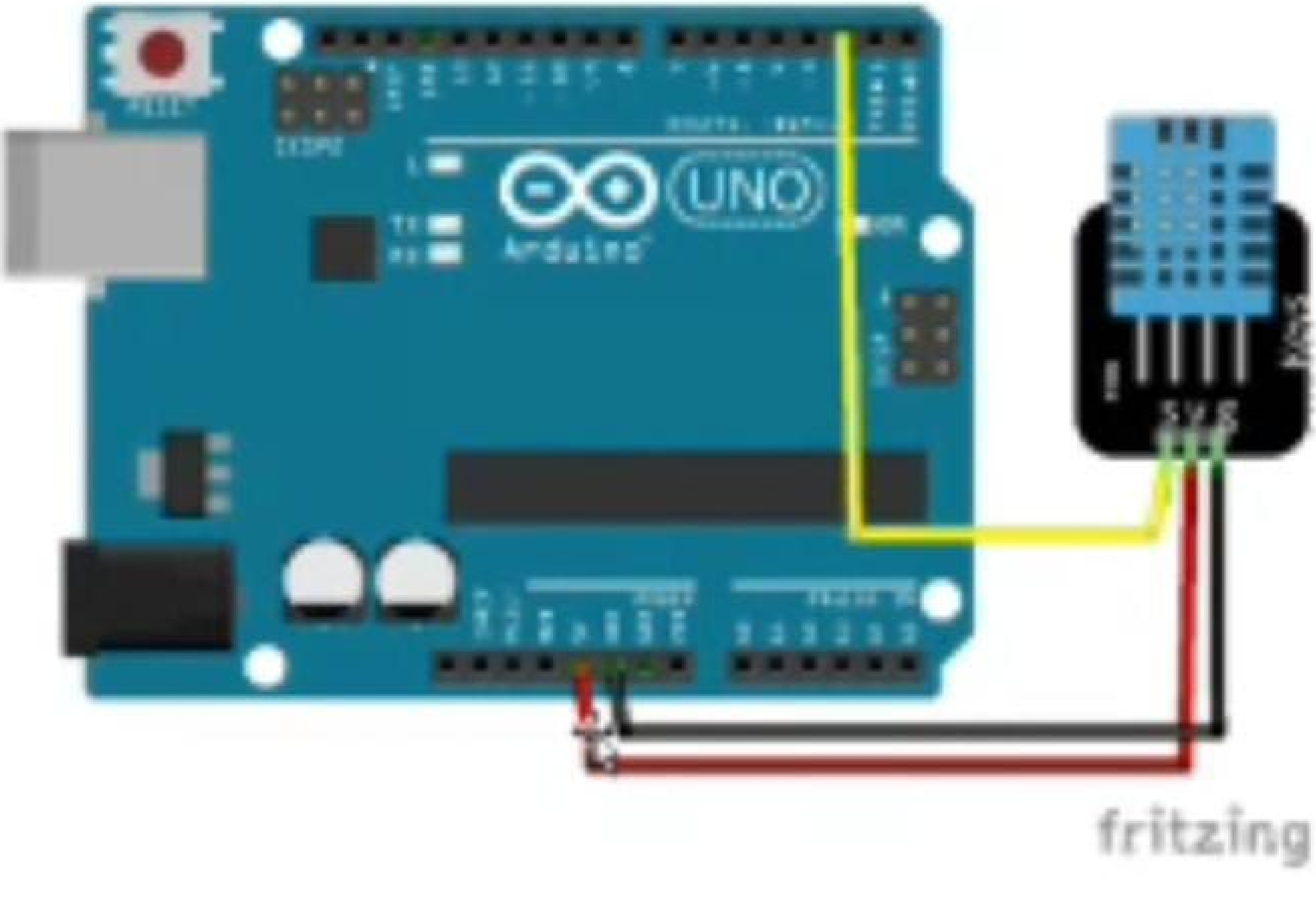
sketch_aug6d.ino

```
int pinDHT11=7;
SimpleDHT11 dht11(pinDHT11);

void setup() {
  // put your setup code here, to run once:
lcd.begin();
lcd.backlight();
}

void loop() {
  // put your main code here, to run repeatedly:
byte temperature;
byte humidity;
dht.read(&temperature,&humidity,NULL);

lcd.setCursor(0,0);
lcd.print("Temperature:");
lcd.print(temperature);
lcd.print("°");  //alt 248

lcd.setCursor(0,1)

```

Activate Windows
Go to Settings to activate Windows.

Building sketch

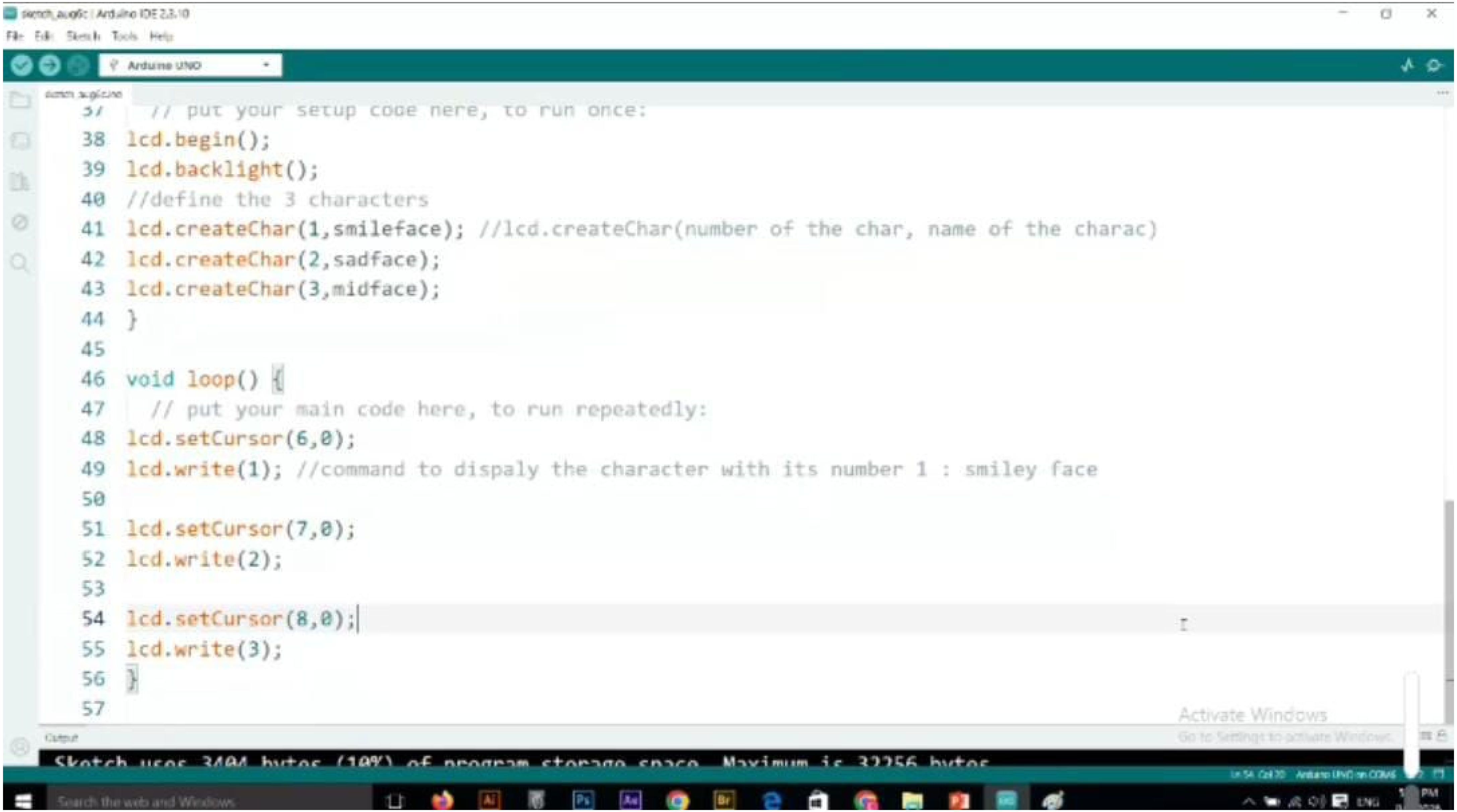
Ln 21, Col 4 Arduino Uno on COM3 [not connected]

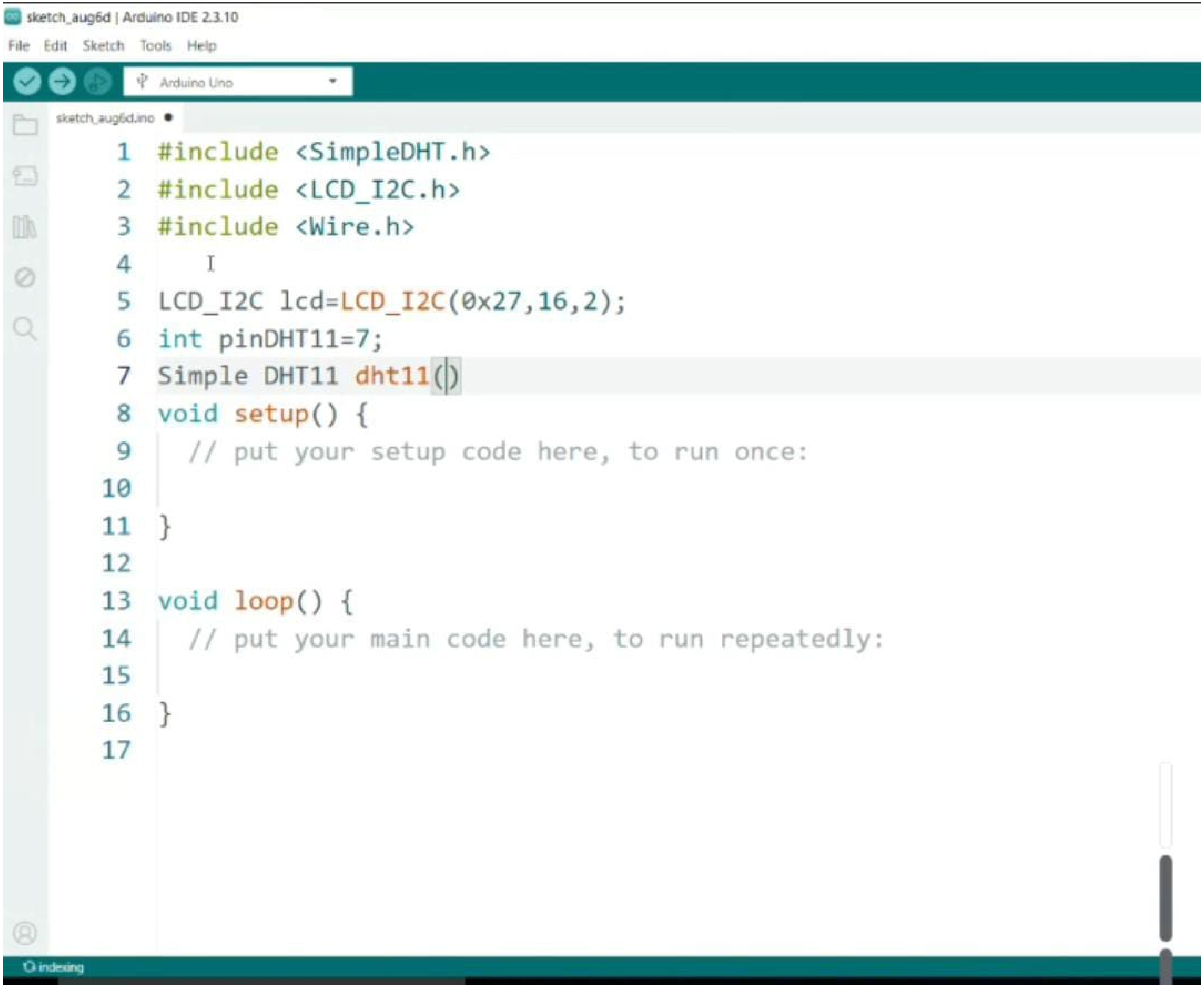

UNO
Arduino
fritzing

byte [8]{

0b 00000,
0b 01010,
0b 01010,
0b 00000,
00100,
10001, 01110, 0000,
01110, 10001, 1111,
0b 00000,

}


File Edit Sketch Tools Help
Arduino UNO
// put your setup code here, to run once:
lcd.begin();
lcd.backlight();
//define the 3 characters
lcd.createChar(1,smileface); //lcd.createChar(number of the char, name of the charac)
lcd.createChar(2,sadface);
lcd.createChar(3,midface);
}
void loop() {
// put your main code here, to run repeatedly:
lcd.setCursor(6,0);
lcd.write(1); //command to dispaly the character with its number 1 : smiley face
lcd.setCursor(7,0);
lcd.write(2);
lcd.setCursor(8,0);
lcd.write(3);
}
Activate Windows
Output
Sketch uses 3404 bytes (10%) of program storage space. Maximum is 32256 bytes.
Search the web and Windows


sketch_aug6d | Arduino IDE 2.3.10
File Edit Sketch Tools Help
Arduino Uno
sketch_aug6d.ino
#include <SimpleDHT.h>
#include <LCD_I2C.h>
#include <Wire.h>
LCD_I2C lcd=LCD_I2C(0x27,16,2);
int pinDHT11=7;
Simple DHT11 dht11()
void setup() {
// put your setup code here, to run once:
}
void loop() {
// put your main code here, to run repeatedly:
}
indexing

I2C_Scanner_Code__1_.ino

```
#include <Wire.h>
#include <LCD_I2C.h>
void setup()
{
Wire.begin();
Serial.begin(9600);
Serial.println("\nI2C Scanner");
}
void loop()
{
byte error, address;
int Devices;
Serial.println("Scanning...");
Devices = 0;
for(address = 1; address < 127; address++ )
{

Wire.beginTransmission(address);
error = Wire.endTransmission();
if (error == 0)
{
Serial.print("I2C device found at address 0x");
if (address<16)
Serial.print("0");
Serial.print(address,HEX);
Serial.println("  !");
Devices++;
}
```

I2C_Scanner_Code__1_.ino

```
{

Wire.beginTransmission(address);
error = Wire.endTransmission();
if (error == 0)
{
Serial.print("I2C device found at address 0x");
if (address<16)
Serial.print("0");
Serial.print(address,HEX);
Serial.println("  !");
Devices++;
}
else if (error==4)
{
Serial.print("Unknown error at address 0x");
if (address<16)
Serial.print("0");
Serial.println(address,HEX);
}
}
if (Devices == 0)
Serial.println("No I2C devices found\n");
else
Serial.println("done\n");
delay(5000);
}

```

Address_scanning__1_.ino

```
#include <Wire.h>

void setup() {
  // put your setup code here, to run once:
  Serial.begin(9600);
while(!Serial){}
Serial.println();
Serial.println("I2C scanner.Scanning...");
byte count=0;
Wire.begin();
for(byte i =8; i<2120; i++){
  Wire.beginTransmission(i);
  if(Wire.endTransmission()==0)
  {
    Serial.print("Found Address:");
    Serial.print(i,DEC);
    Serial.print("(0x");
    Serial.print(i,HEX);
    Serial.println(")");
    count++;
    delay(1);
}
}
Serial.println("Done");
Serial.print("Found");
Serial.print(count,DEC);
Serial.println("device(s).");
}
```

Address_scanning__1_.ino

```
Serial.println();
Serial.println("I2C scanner.Scanning...");
byte count=0;
Wire.begin();
for(byte i =8; i<2120; i++){
  Wire.beginTransmission(i);
  if(Wire.endTransmission()==0)
  {
    Serial.print("Found Address:");
    Serial.print(i,DEC);
    Serial.print("(0x");
    Serial.print(i,HEX);
    Serial.println(")");
    count++;
    delay(1);
}
}
Serial.println("Done");
Serial.print("Found");
Serial.print(count,DEC);
Serial.println("device(s).");
}

void loop() {
  // put your main code here, to run repeatedly:

}
```

Arduino LCD I2C & DHT11 Code Snippets

1. Basic LCD Hello World

```
#include <Wire.h>
#include <LCD_I2C.h>

//Define the LCD
LCD_I2C lcd=LCD_I2C(0x27,16,2); //0x27 : code for using the I2C scanner ,16 : number of columns, 2: number of rows

void setup() {
  // put your setup code here, to run once:
  lcd.begin(); // command to initialize the LCD
  lcd.backlight(); //command to turn on the LCD Blue backlight
}

void loop() {
  // put your main code here, to run repeatedly:
  lcd.setCursor(1,0); // command will set the cursor on the second column (1) and the first row (0)
  lcd.print("Hello Arduino !");
  lcd.setCursor(2,1); //command will set the cursor on the third column (2) and the second row(1)
  lcd.print("LCD Tutorial");
}
```

2. Autoscroll - Welcome to Arduino LCD Tutorial

```
#include <LCD_I2C.h>
#include <Wire.h>

LCD_I2C lcd=LCD_I2C(0x27,16,2);

void setup() {
  // put your setup code here, to run once:
  lcd.begin();
  lcd.backlight();
}

void loop() {
  // put your main code here, to run repeatedly:
  lcd.setCursor(15,0);
  lcd.autoscroll();

  String sentence="Welcome to Arduino LCD Tutorial";
  for(int i=0;i<31;i++){ //31 characters in the sentence including the space
    lcd.print(sentence[i]);
    delay(500);
  }
}
```

3. Autoscroll - Welcome to Arduino World

```
#include <LCD_I2C.h>
#include <Wire.h>

LCD_I2C lcd=LCD_I2C(0x27,16,2);

void setup() {
  // put your setup code here, to run once:
  lcd.begin();
  lcd.backlight();
}

void loop() {
  // put your main code here, to run repeatedly:
```

```
  lcd.setCursor(16,0);
  lcd.autoscroll();

  String sentence="Welcome to Arduino World";
  for(int i=0;i<=23;i++){ //23 characters in the sentence including the space
    lcd.print(sentence[i]);
    delay(500);
  }
}
```

4. Custom Characters - Static Display

```
#include <LCD_I2C.h>
#include <Wire.h>

LCD_I2C lcd=LCD_I2C(0x27,16,2);

byte smileface[8]={
  0b00000,
  0b01010,
  0b01010,
  0b00000,
  0b10001,
  0b01110,
  0b00000,
};

byte sadface[8]={
  0b00000,
  0b01010,
  0b01010,
  0b00000,
  0b01110,
  0b10001,
  0b00000,
};

byte midface[8]={
  0b00000,
  0b01010,
  0b01010,
  0b00000,
  0b00000,
  0b11111,
  0b00000,
};

void setup() {
  // put your setup code here, to run once:
  lcd.begin();
  lcd.backlight();
  //define the 3 characters
  lcd.createChar(1,smileface); //lcd.createChar(number of the char, name of the charac)
  lcd.createChar(2,sadface);
  lcd.createChar(3,midface);
}

void loop() {
  // put your main code here, to run repeatedly:
  lcd.setCursor(6,0);
  lcd.write(1); //command to dispaly the character with its number 1 : smiley face

  lcd.setCursor(7,0);
  lcd.write(2);

  lcd.setCursor(8,0);
  lcd.write(3);
}
```

5. Custom Characters - Animated (Cycling Faces)

```
#include <LCD_I2C.h>
#include <Wire.h>

LCD_I2C lcd=LCD_I2C(0x27,16,2);

byte smileface[8]={
  0b00000,
  0b01010,
  0b01010,
  0b00000,
  0b10001,
  0b01110,
  0b00000,
};

byte sadface[8]={
  0b00000,
  0b01010,
  0b01010,
  0b00000,
  0b01110,
  0b10001,
  0b00000,
};

byte midface[8]={
  0b00000,
  0b01010,
  0b01010,
  0b00000,
  0b00000,
  0b11111,
  0b00000,
};

void setup() {
  // put your setup code here, to run once:
  lcd.begin();
  lcd.backlight();
  //define the 3 characters
  lcd.createChar(1,smileface); //lcd.createChar(number of the char, name of the charac)
  lcd.createChar(2,sadface);
  lcd.createChar(3,midface);
}

void loop() {
  // put your main code here, to run repeatedly:
  lcd.setCursor(7,0);
  lcd.write(1); //command to dispaly the character with its number 1 : smiley face
  delay(1000);

  lcd.setCursor(7,0);
  lcd.write(2);
  delay(1000);

  lcd.setCursor(7,0);
  lcd.write(3);
  delay(1000);
}
```

6. DHT11 Temperature & Humidity (LCD_I2C library)

```
#include <SimpleDHT.h>
#include <LCD_I2C.h>
#include <Wire.h>
//DEFINE THE LCD
```

```
LCD_I2C lcd=LCD_I2C(0x27,16,2);
//DEFINE THE DHT SENSOR
int pinDHT11=7;
SimpleDHT11 dht11(pinDHT11);

void setup() {
  // put your setup code here, to run once:
  lcd.begin();
  lcd.backlight();
}

void loop() {
  // put your main code here, to run repeatedly:
  byte temperature;
  byte humidity;
  //to read the temp and humidity from the sensor
  dht11.read(&temperature,&humidity,NULL);

  lcd.setCursor(0,0);
  lcd.print("Temp:");
  lcd.print(temperature);
  lcd.print(" degrees");  //alt 248

  lcd.setCursor(0,1);
  lcd.print("Humidity:");
  lcd.print(humidity);
  lcd.print("%");

  delay(2000);
}
```

7. DHT11 Temperature & Humidity (LiquidCrystal_I2C library)

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <SimpleDHT.h>

int pinDHT11 = 2;
SimpleDHT11 dht11(pinDHT11);
LiquidCrystal_I2C lcd(0x27, 16, 2);

void setup() {
  lcd.begin(16, 2);
  lcd.backlight();
}

void loop() {
  byte temperature = 0;
  byte humidity = 0;

  if (dht11.read(&temperature, &humidity, NULL) != SimpleDHTErrSuccess) {
    lcd.setCursor(0, 0);
    lcd.print("Read Error");
    delay(1000);
    return;
  }

  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("Temp: ");
  lcd.print((int)temperature);
  lcd.print((char)223);
  lcd.setCursor(0, 1);
  lcd.print("Humidity: ");
  lcd.print((int)humidity);
  lcd.print("%");

  delay(2000);
```

```
}
```

8. DHT11 Temperature & Humidity (Degree Symbol Variant)

```
#include <SimpleDHT.h>
#include <LCD_I2C.h>
#include <Wire.h>

//DEFINE THE LCD
LCD_I2C lcd=LCD_I2C(0x27,16,2);
//DEFINE THE DHT SENSOR
int pinDHT11=7;
SimpleDHT11 dht11(pinDHT11);

void setup() {
  // put your setup code here, to run once:
  lcd.begin();
  lcd.backlight();
}

void loop() {
  // put your main code here, to run repeatedly:
  byte temperature;
  byte humidity;
  //to read the temp and humidity from the sensor
  dht11.read(&temperature,&humidity,NULL);

  lcd.setCursor(0,0);
  lcd.print("Temperature:");
  lcd.print(temperature);
  lcd.print((char)223);

  lcd.setCursor(0,1);
  lcd.print("Humidity:");
  lcd.print(humidity);
  lcd.print("%");

  delay(2000);
}
```